Spatial–Spectral–Semantic Cross-Domain Few-Shot Learning for Hyperspectral Image Classification

Mengxin Cao, *Member, IEEE*, Xu Zhang, Jinyong Cheng, *Member, IEEE*, Guixin Zhao, Wei Li, *Senior Member, IEEE*, and Xiangjun Dong

***Abstract*— Preprocessing procedures are commonly employed to reduce water-absorption bands and noise in hyperspectral images (HSIs). Nevertheless, they typically do not entirely eradicate noise. This is especially evident in scenarios that necessitate data of exceptional quality, such as cross-domain few-shot classification tasks. Within these specific conditions, the influence of remaining background noise on the ultimate results of classification is substantial. Furthermore, the presence of sample selection biases in the few-shot task might lead to the emergence of false statistical correlations between data from distinct domains, resulting in a decrease in the model's ability to generalize. We propose a new method called spatial–spectral–semantic cross-domain few-shot learning (S^3CFSL) to address the challenge. This method promotes the learning of transferable information by incorporating feature denoising operations in the feature extraction process to restore essential information. Concurrently, it enhances cross-domain distributional consistency by introducing a semantic-aware strategy to strengthen the association between cross-domain data and semantic information. Specifically, the spatial and spectral dual channels (SSDCs), in conjunction with the cross-spatial–spectral transformer (CSST), are designed as a feature extractor to acquire interactive spatial–spectral features. The feature-denoising operations can further acquiring transferable information from cross-domain features, thus facilitating meta-learning in both the source domain (SD) and the target domain (TD). Meanwhile, a semantic-enhanced domain alignment (SEDA) is designed to promote domain adaptation by using a semantic-aware strategy, which significantly enhances distributional consistency for cross-domain tasks. Our results exhibit exceptional classification efficacy in comparison to other state-of-the-art approaches on three public HSI datasets.**

***Index Terms*— Cross-domain, distribution alignment, domain adaption, few-shot learning (FSL), hyperspectral image (HSI) classification.**

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Mengxin Cao, Xu Zhang, Jinyong Cheng, Guixin Zhao, and Xiangjun Dong are with the Key Laboratory of Computing Power Network and Information Security, Ministry of Education, Shandong Computer Science Center, Qilu University of Technology (Shandong Academy of Sciences), Jinan 250316, China, and also with Shandong Provincial Key Laboratory of Computer Networks, Shandong Fundamental Research Center for Computer Science, Jinan 250014, China (e-mail: zgx@qlu.edu.cn; dxj@qlu.edu.cn).

Wei Li is with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: liwei089@ieee.org).

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I. Introduction

HYPERSPECTRAL image (HSI) is composed of numerous small spectral bands, which can be represented as a 3-D data cube [1]. This structure enables HSI to obtain important information through rich spectral representation. In recent years, HSI has received a lot of attention for related applications such as medicine [2], soil property prediction [3], and multispectral fusion [4]. HSI classification as a hot HSI task aims to assign a specific label to each pixel [5]. The initial investigations mostly concentrated on direct classification utilizing fundamental pattern recognition techniques such as K-nearest neighbors (K-NNs) [6] and support vector machines (SVMs) [7]. Recently, deep learning has significantly improved the classification ability of HSI by extracting spatial–spectral features through powerful feature representations like convolutional neural network (CNN) [8], [9], group convolutional network (GCN) [10], [11], recurrent neural network (RNN) [12], etc. For better representation of spectral long-distance features [13], Transformers have been applied to HSI tasks because of their effective global feature extraction abilities [14], [15]. The attention mechanism in Transformer enables good modeling of the global features of HSI data, which are used to emphasize useful spectral information [16], [17]. However, training deep learning models usually requires a large amount of labeled data. Acquiring labeled samples for HSI is both time-consuming and requires a lot of effort. In practical HSI classification applications, the distribution of labeled data is often imbalanced [18]. The target domain (TD) refers to HSI with a limited amount of labeled data, while the source domain (SD) refers to HSI with a plentiful amount of labeled data [19]. HSI cross-domain classification refers to the process of utilizing SD to acquire an adequate number of labeled samples for the purpose of classifying TD [20]. Nevertheless, the task of acquiring the common knowledge across the SD and TD and utilizing effective transferable knowledge to attain efficient classification performance continues to be a difficult undertaking [21].

Currently, for such cross-domain HSI tasks, scholars have made a series of efforts, and the current work can be roughly categorized into three groups: distribution alignment [22], [23], feature selection-based methods [24], [25], and subspace learning-based methods [26], [27]. For distribution alignment methods, scholars strive to create a model that minimizes the difference between the distributions of the SD and the TD [28].

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The feature selection-based method focuses on extracting related features from diverse domains and establishing a shared feature space [29]. For subspace learning-based methods, the distribution gap between the SD and the TD is aligned by projecting HSI data from both domains into a common subspace [30]. Each of these solutions has the ability to partially address the domain shift problem in the cross-domain task. However, the typical approach of decreasing the shift between two domains is predicated on the assumption that the classes of the SD and the TD are identical. The aforementioned methods pose challenges in resolving the presence of new classes in the TD for categorization [31].

As a meta-learning method, few-shot learning (FSL) is able to quickly build the cognitive ability to new concepts with just one or a few examples to gain the ability to predict the target class data [32], [33]. It employs labeled source class in a set-based approach to simulate few-shot tasks [34]. After the training process, the meta-knowledge is stored in a model. Subsequently, this model is employed to classify a new FSL task by using a limited collection of labeled target class data and meta-knowledge and predict labels for all unlabeled target class data [35]. Based on FSL methods, researchers have also attempted to apply supervised contrast learning methods to cross-domain FSL to promote intraclass compactness and interclass discretization of features in metric space [34], [36].

Although these networks have gained a promising boost on cross-domain few-shot problems, they still have not focused on optimizing the original data and modeling semantic relationships. The particular constraints can be generally outlined as follows.

1) Currently, most strategies for cross-domain few-shot classification of HSI focus on optimizing the direct measurement distances of samples inside the meta-migration framework of FSL [37]. However, these methods tend to ignore the role of improving the representation of essential features. Some bands inevitably have low signal-to-noise ratios due to limitations in the sensors, ambient conditions, and atmospheric impacts [38]. Although the water-absorbing and noise bands have been removed, residual bands often still contain some level of noise [39], [40]. Relying only on deep learning methods is not effective enough to exclude the effects of different pixels and various noises on feature extraction [41]. Furthermore, the noise generated under multiple layers of extraction and activation functions of deep learning networks exacerbates the effect of spectral aberration [42]. Reducing the number of feature extraction layers and adding feature denoising steps not only ensures the adequate extraction of spatial–spectral features but also mitigates the impact of non-essential features on transferable knowledge in meta-learning.
2) Furthermore, the process of sample selection is marked by significant randomness, resulting in considerable spatial heterogeneity within the limited sample size. The presence of sample selection bias in cross-domain few-shot classification tasks can lead to imprecise statistical associations between class meanings, which might hinder the model's ability to generalize effectively [43]. Adding semantic masks enables the discrimination of the extraction features, which in turn facilitates more effective domain alignment [44], [45]. The semantic-aware strategy can be further planned to enhance the separation between different classes by clustering of semantic information, hence mitigating domain shift.

To address the above problems, we propose a spatial–spectral–semantic cross-domain FSL (S^3CFSL) method. S^3CFSL further mitigates the effect of the activation functions and noise on features by using an efficient feature extraction network and feature denoising operations to improve transferable learning capabilities in meta-learning. Simultaneous use of semantic-aware strategy enhances semantic associations, thereby facilitating domain alignment. Specifically, in the first stage, we establish an FSL task for both SD and TD. This phase encompasses three key steps. First, features within the SD and TD are aligned to a uniform dimension. Subsequently, these features undergo processing via the cross spatial–spectral Transformer (CSST), employing spatial and spectral dual-channels (SSDCs), which can capture fused spatial–spectral features. Finally, it involves executing feature denoising operations, which are crucial to reduce noise and get the essential features. In the second stage, after the completion of the FSL, semantic-enhanced domain alignment (SEDA) uses the semantic-aware strategy to aggregate samples and promotes domain adversarial adaptive capability through adding semantic masks, ultimately achieving domain alignment.

The main contributions of this work are summarized as follows.

1) Starting from the problems of the influence of sample external factors and the uneven distribution of classes in cross-domain problems, we propose a novel cross-domain FSL method called S^3CFSL. It can efficiently accomplish the HSI classification task by capturing sufficient transferable information and designing the semantic-aware strategy for domain alignment.
2) We design CSST with SSDC as an embedding to reduce the number of layers in order to fusion interactive spatial and spectral features.
3) Feature denoising operations are introduced in the HSI cross-domain FSL task to reduce the impact of external and internal noise, which improves the transferable learning for meta-learning.
4) The design of SEDA aims to facilitate the method's domain alignment capability. It introduces the semantic-aware strategy to utilize semantic information, which finally mitigates cross-domain distributional differences.

A preliminary version of this article can be found in [46]. The new contributions arise from four primary facets. Initially, improvements are made to the process of extracting features. The CSST with SSDC is developed to acquire fused spatial–spectral information. Second, the incorporation of feature denoising procedures serves to highlight crucial information while diminishing the noise band and the influence of deep learning multilayer extraction. This method additionally aids in the development of transferable knowledge. Third, effectively addresses the issue of spatial distribution disparities in the cross-domain problem by employing

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semantic information and implementing an idea that is conscious of semantic context. Finally, we validate the proposed method on other publicly available datasets. In the ablation study, we added the validation of the newly proposed module.

The remainder of this article is structured as follows. In Section II, we delve into recent advances in domain adaptation in HSI classification and the application of Transformer to computer vision (CV). Section III outlines the detailed implementation of the proposed S^{3}CFSL, and Section IV presents the experimental results obtained from applying this method across three distinct target datasets. Finally, Section V encapsulates our conclusion.

II. Related Work

A. Cross-Domain FSL

Several meta-learning-based FSL methods exhibit suboptimal performance in cross-domain scenarios, particularly when there is a significant domain shift between a base class and a new class. A cross-domain FSL method has been suggested to tackle this problem. The process entails extracting base and new classes from separate domains, where the class label sets are non-overlapping. The HSI dataset is divided into SD and TD. The SD consists of a substantial number of labeled samples, whereas the TD has only a few labeled samples, which allows for the transfer of knowledge from SD to TD. This introduces two types of cross-domain FSL tasks: the SD FSL task and the TD FSL task. The two task processes are carried out alternately, enabling the embedded feature extractor to acquire knowledge on how to establish a connection between SD and TD within a common feature space. Li et al. [31] introduced a deep cross-domain FSL method in the HSI classification sector. Zhang et al. [21] developed the graph information aggregation CFSL (GiaCFSL) approach. Wang et al. [47] designed a two-branch domain adaptive FSL network for domain fusion and domain separation to achieve independent category distribution, respectively.

These works signify notable advancements in HSI classification. However, these methods still overlook the improvement of sample quality and the alignment of spatial differences.

B. Transformer Module in CV

In CV tasks, Sun et al. [5] presented the vision Transformer (ViT) model for CV problems, integrating the Transformer encoder into image classification. In ViT, images are first flattened into 2-D patches. Patch embedding standardizes the self-attention mechanism's input sequence dimensions. The classification result is then obtained using a multilayer perceptron (MLP) module. The whole structure is composed of the multihead self-attention (MSA) mechanism, as shown in Fig. 1.

The transformer has become increasingly prominent and successful in the field of HSI classification in recent years, primarily due to its robust capability to describe spectral long distances [48], [49]. In the HSI classification task, He et al. [50] introduced T-SST-L in the field of HSI classification. This model integrates a CNN to extract spatial features, together with a modified Transformer to capture sequential spectral connections. The method proposed in [13] utilizes densely sampled data to divide the spectral channel into various groups. It utilized groupwise spectral embedding methods and a Transformer encoder module to extract sophisticated features.

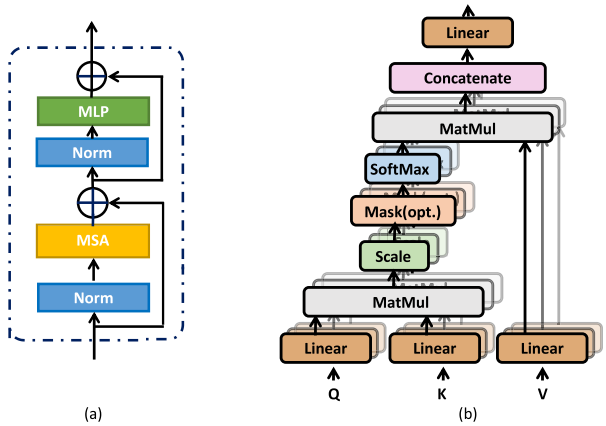


Fig. 1. Structure of the Transformer. (a) Overall pipeline of the transformer block. (b) Structure of MSA.

For the feature extraction section, we propose a new spatial–spectral Transformer structure called CSST, which performs feature embedding through SSDC and then combines the cross-attention part for deeper information extraction.

III. Proposed Method

A. Overall Architecture

The proposed S^{3}CFSL in the training phase is shown in Fig. 2. The process encompasses two primary stages: FSL and domain alignment. Within the FSL stage, two predominant types of FSL are identified. One is the FSL for the SD. The other is the FSL for the TD. These two types of FSLs are alternated. There are three main parts for each type of FSL. First, mapping operations are performed on the dimensions of the SD and TD datasets so that they have the same dimension value (for example, if the smallest dimension in the source and TD datasets is 100, set 100 as the uniform dimension). Second, the SD and TD datasets are input into the SSDC to perform the extraction and embedding of spatial and spectral features and into the CSST feature extractor to obtain interactive spatial–spectral features. Third, feature denoising operations on the retrieved features not only reduce the noise but also mitigate the effect of multilayer activation functions. After these three feature processing processes, we get spatial–spectral embedded features. We then evaluate the distances between unlabeled and labeled samples for each class in the spatial–spectral embedding to perform SD or TD FSL. Then, obtain the classification loss L_{fsl}^{s} and L_{fsl}^{t} for all query samples in the SD and TD, respectively. In the domain alignment stage, SEDA is designed to mitigate domain shift, rendering the extracted spatial–spectral embedded features domain-invariant. This approach facilitates domain alignment and results in the domain alignment loss, denoted as L_{DA}.

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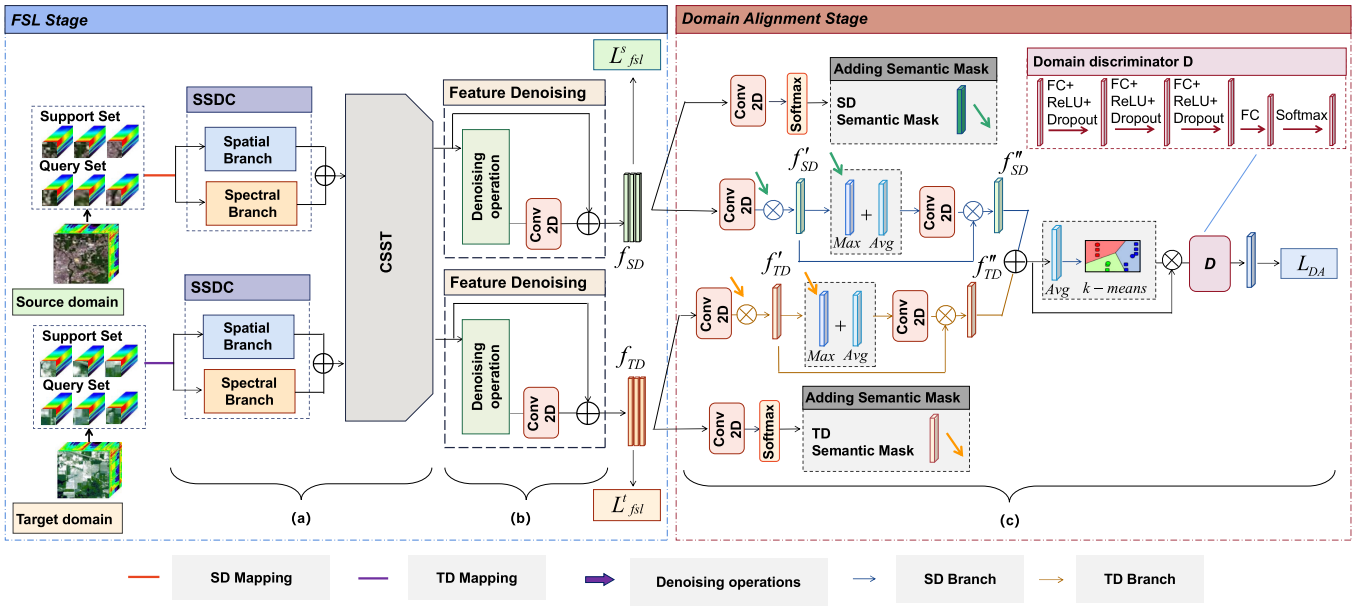


Fig. 2. Diagram of the proposed S^{3}CFSL method. (a) Feature extraction. (b) Feature denoising. (c) SEDA.

Therefore, the total loss function of the SD and TD FSLs with domain alignment is denoted as L of S^{3}CFSL, and the process can be described as

$$\begin{aligned} L^s &= L^s_{\text{fsl}} + L_{\text{DA}} \\ L^t &= L^t_{\text{fsl}} + L_{\text{DA}} \\ L &= L^s + L^t. \end{aligned} \tag{1}$$

In addition, the testing phase includes four steps. First, the dimensions are mapped uniformly. Second, spatial–spectral embedded features are extracted via SSDC and CSST. Third, denoising operations are performed on the features. Finally, unlabeled samples are classified via a nearest neighbor (NN) classifier. After training, the CSST model produces discriminative features with low intraclass and significant interclass variability. Thus, the NN classifier excels in classifying unlabeled TD samples. Eventually, classification maps for all TD samples are created to evaluate S^{3}CFSL's performance.

B. Cross-Domain FSL

Meta-learning simulates a few-shot classification task with variable land cover in the TD for HSI classification. It builds a few-shot task to extract relevant knowledge for classification. The FSL creates a support set $\text{SP}_S = \{(x_i, y_i)\}_{i=1}^{C_s \times N}$ and a query set $\text{QU}_S = \{(x_i, y_i)\}_{i=1}^{C_s \times M}$ for the SD, while creating a support set $\text{SP}_T = \{(x_i, y_i)\}_{i=1}^{C_t \times N}$ and a query set $\text{QU}_T = \{(x_i, y_i)\}_{i=1}^{C_t \times M}$ for the TD. We select C classes from the dataset randomly and extract N and M labeled samples for each class to form support and query sets, respectively. Therefore, we use a meta-learning-based few-shot classification method that can solve the problem of class inconsistency between the SD and TD. Specifically, in the training phase, the network parameters are updated using a non-parametric softmax based on the results of the distance metric. The class distribution for a query sample x_j in the query set QU is

$$P\left(y_j = n \mid x_j \in \text{QU}\right) = \frac{\exp\left(-d\left(\text{H}\left(T_\varphi\left(x_j\right)\right), c_n\right)\right)}{\sum_{n=1}^{C} \exp\left(-d\left(\text{H}\left(T_\varphi\left(x_j\right)\right), c_n\right)\right)} \tag{2}$$

where QU is the query set, $d(\cdot)$ denotes a Euclidean distance function, c_n is the embedded features of the nth class in the support set, $\text{H}(\cdot)$ denotes the feature denoising operations, $T_\varphi(\cdot)$ denotes the feature extractor network with parameter φ, x_j is the sample of the query set, and y_j is the label of x_j.

The FSL loss for all query samples in an SD query sample x and its true class label n can be calculated as follows:

$$L^s_{\text{fsl}} = E_{\text{SP}_S, \text{QU}_S}\left[-\sum_{(x,y)\in \text{QU}_S} \log \text{H}\left(f_\varphi(y = n \mid x)\right)\right] \tag{3}$$

where SP_S and QU_S are the support and query sets from the SD dataset, respectively.

By a similar process, we obtain the loss of the FSL in TD L^t_{fsl}.

We have developed the CSST model with a less hierarchical structure specifically designed for efficient meta-learning. This model ensures the extraction of sufficient spatial and spectral features while minimizing the distortion of feature details. The feature extraction process of the SSDC is integrated into the CSST model. In addition, our feature extraction approach is used in the succeeding feature denoising operations to reduce the influence of noise features on transferred information. The purpose of this dual-stage strategy is to improve the extraction of essential features, thereby enabling learning from a few samples.

1) Features Extraction: In order to extract interactive spatial–spectral features and obtain essential features, we designed SSDC with the CSST feature extraction process.

In the feature extraction stage, the input HSI data are first embedded into the same dimensions and fed into the SSDC

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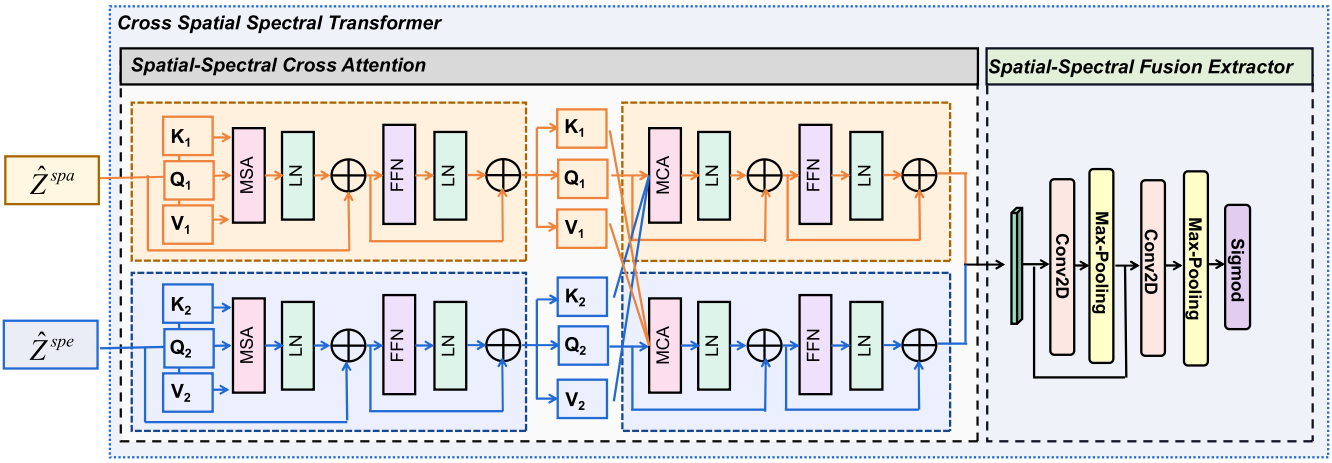


Fig. 3. Architecture of the CSST.

for spectral and spatial feature extraction, respectively. The obtained spectral and spatial features are input into the CSST, and the fully extracted spatial–spectral features are obtained by the spatial–spectral cross-attention (SSC Attention) mechanism of design.

a) Spatial and spectral dual-channel: To enhance the extraction of spatial and spectral features, the HSI is initially subjected to dual-channel processing for feature embedding. Specifically, we divide the HSI into patches of size 9×9 and input them into the spatial and spectral embedding layers, respectively. The process can be described as follows:

$$\begin{aligned} \hat{Z}^1 &= \text{Concat}(\text{Conv}(Z), \text{Sobel}(Z)) \\ \hat{Z}^2 &= \text{Concat}(\text{Conv}(Z), \text{Sigm}(Z)) \end{aligned} \tag{4}$$

where $\text{Concat}(\cdot)$ refers to concatenation in the channel dimension, the $\text{Conv}(\cdot)$ denotes convolution, $\text{Sigm}(\cdot)$ denotes sigmoid function, and $\text{Sobel}(\cdot)$ denotes Sobel operation, respectively.

The spatial embedded feature $\hat{Z}^1$ and the spectral embedded feature $\hat{Z}^2$ are obtained, respectively. Then the $\hat{Z}^1$ and $\hat{Z}^2$ are fed into the CSST block for further feature extraction.

b) Cross-spatial–spectral transformer: Recently, deep learning methods based on feature fusion have been developing rapidly in the field of CV [51], [52]. Meanwhile, through efficient special fusion, the fully fused visual representation information is extracted, and the subsequent tasks can be carried out more effectively.

Therefore, we design the CSST to fusion and enhance the extraction of spatial–spectral features. As shown in Fig. 3, our Transformer consists of an SSC attention and a spatial–spectral fusion extractor (SSFE).

We have designed SSC attention to perform feature extraction. It uses MSA and multihead cross-attention (MCA) for information interaction and feature extraction performance. The MSA for deep extraction of spatial and spectral features and MCA to achieve global context exchange across domains for information interaction in spatial and spectral features, as shown in Fig. 3.

The key component of the Transformer is the multihead attention block. The input feature representations are converted into three linear projections with dimension, conventionally named query($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$), as the input of the attention layers. The attention operation is denoted as

$$\text{Attention}\ (\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right) \cdot \mathbf{V} \tag{5}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ denote query ($\mathbf{Q} = [q_1, \ldots, q_m]$), key ($\mathbf{K} = [k_1, \ldots, k_m]$), and value ($\mathbf{V} = [v_1, \ldots, v_m]$), respectively. Using (5), several different self-attention layers can be assembled into one MSA. Following this, a feed-forward network (FFN) is employed, comprising two MLP layers with a GELU activation function. The primary purpose of this FFN is to further refine the feature tokens generated by the MSA mechanism.

After the MSA operations, the cross-attention mechanism further fuses the extracted spatial–spectral features. Unlike the MSA, the MCA stage is capable of fusing features between different blocks and deep interaction to obtain the spatial $\mathbf{Q}_1', \mathbf{K}_1', \mathbf{V}_1'$ and spectral $\mathbf{Q}_2', \mathbf{K}_2', \mathbf{V}_2'$ with fusion features, respectively. The specific process is defined as

$$\begin{aligned} \{\mathbf{Q}_1', \mathbf{K}_1', \mathbf{V}_1'\} &= \left\{ Z^1\mathbf{W}_1^{Q_1\prime}, Z^1\mathbf{W}_1^{K_1\prime}, Z^1\mathbf{W}_1^{V_1\prime} \right\} \\ \{\mathbf{Q}_2', \mathbf{K}_2', \mathbf{V}_2'\} &= \left\{ Z^2\mathbf{W}_2^{Q_2\prime}, Z^2\mathbf{W}_2^{K_2\prime}, Z^2\mathbf{W}_2^{V_2\prime} \right\}. \end{aligned} \tag{6}$$

Subsequently, the extracted features are also processed in the FFN.

Finally, we perform a simple concatenation operation of the spatial and spectral features that have been interacted with and fully extracted with the features and input them into SSFE for further feature fusion.

A simple post-connection spatial–spectral feature obtained after cross-attention. However, this shallow fusion cannot meet the feature extraction requirements for small samples.

We deploy SSFE to aggregate spatial and spectral features in order to fuse spatial–spectral features further. Given the shallow spatial–spectral features Z^f from SSC attention, the SSFE can be formulated as follows:

$$\hat{Z}^f = \text{Sigm}\big(\text{Max}\big(\text{Conv}\big(Z^f + \text{pool}\big(\text{Conv}\big(Z^f\big)\big)\big)\big)\big) \tag{7}$$

where $\text{Max}(\cdot)$ stand for max-pooling operations.

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After the complete feature extraction process, we get $\hat{Z}^f$, the deeply extracted essential features for further feature denoising operations.

2) Feature Denoising: In order to obtain high-quality features and reduce the influence of external factors on learning transferable features, we add a feature denoising operation.

For the feature denoising operations, we use a Gaussian filter to highlight the high-quality feature information of the labeled pixels in the center region [53]. Gaussian filter is a linear smoothing filter that suppresses high-frequency noise by weighted averaging, and the weights of the features decrease as the distance from the center region increases. Let the coordinates of the pixel (0, 0) in the window be a and b, and the standard deviation be σ. The weight of the pixel (0, 0) in the center of the window is defined by a 2-D Gaussian function as

$$\mathrm{G}(a,b)=\frac{1}{2\pi\sigma^2}e^{-\frac{a^2+b^2}{2\sigma^2}}. \tag{8}$$

The Gaussian denoised features are merged with the features that are not subjected to the denoising operations to obtain f. This is done to prevent excessive blurring of the spatial features, which results in excessive missing information. This is specifically expressed as follows:

$$f=\mathrm{G}\left(\hat{Z}^f\right)+\hat{Z}^f. \tag{9}$$

With this feature denoising operation, the extracted higher-order spatial–spectral features reduce the effects of multilayer activation functions and other noises. The obtained high-quality learnable features can facilitate the meta-learning process.

C. Domain Alignment

To better facilitate domain alignment, we propose SEDA, which is shown in the right part of Fig. 2. The global data distribution in the SD and TD is adjusted using the SEDA. With conditional domain adversarial network [54], we reduce the domain shift by further planning the semantic-aware strategy.

The classifier prediction g contains discriminative information that can be conditioned on the adversarial adaptation of the feature representation f. By conditioning, the domain variance of f and g can be modeled simultaneously. The domain adversarial loss function L is defined on the domain discriminator D with feature representation $f=f(x)$ and classifier prediction $g=g(x)$

$$\begin{aligned}L_{\mathrm{DA}}=\min_{D}\max_{F,G}\; & E_{x_i^s\sim P_s(x)}\log\left[\mathrm{D}\left(f_i^s,g_i^s\right)\right]\\ & +E_{x_j^t\sim P_t(x)}\log\left[1-\mathrm{D}\left(f_j^t,g_j^t\right)\right]\end{aligned} \tag{10}$$

the function $\mathrm{D}(\cdot)$ measures the probability that x is part of the SD sample category, while the function $1-\mathrm{D}(\cdot)$ indicates the probability that x is part of the TD sample category.

To make the domain discriminator D efficient in distinguishing the SD and TD features for adversarial adaptation, we use a semantic-aware strategy to enhance the utilization of semantic information. Specifically, we use semantic feature normalization to approximate the equations. As shown in Fig. 2, we first randomly select the feature dimensions of different domains and generate semantic mask M with domain semantic information through convolutional layers. Taking the semantic perception process of SD features as an example, the process can be expressed as

$$f'_{\mathrm{SD}}=f_{\mathrm{SD}}\otimes M \tag{11}$$

where $\otimes$ denotes Hadamard product and f' represents feature with semantic mask. The generated semantic mask can be too rough to model semantic information accurately. We therefore perform feature further extraction f''_{SD} given by

$$f''_{\mathrm{SD}}=\mathrm{Sigm}\left(\mathrm{Conv}\left(\left[\mathrm{Max}\left(f'_{SD}\right);\mathrm{Avg}\left(f'_{\mathrm{SD}}\right);M\right]\right)\right)\otimes f'_{\mathrm{SD}} \tag{12}$$

where $\mathrm{Avg}(\cdot)$ is represents average-pooling features.

Similarly, the semantic perception process for TD features is the same as above. Afterward, f''_{SD} and f''_{TD} are combined to obtain f_D.

Finally, in order to flexibly identify feature elements, we apply k-mean clustering on the spatial feature maps obtained via average-pooling along the dimension axis. The process can be described as

$$f=I(\mathrm{Avg}(f_D))+f_D \tag{13}$$

where $I(\cdot)$ represents the k-mean clustering operations, and the value of k is set to 2.

After this step, the features of different domains have more obvious semantically discriminative information, which further facilitates the differentiation ability of the domain discriminator. Domain alignment is finally achieved by using the semantic-aware strategy to mitigate domain shift.

IV. Experiments

In the following, we describe the datasets and the experimental setup, including implementation details and experiments with state-of-the-art approaches. Then, we conduct extensive comparative validations to illustrate that the proposed method outperforms other methods. Finally, we conduct extensive ablation analysis experiments to evaluate important designed modules of S^{3}CFSL.

A. Dataset Description

We chose one source and three TD datasets, and their false-color image is displayed in Fig. 4, and the specific dataset selection is displayed in Tables I–IV.

1) SD Dataset: The Chikusei dataset was gathered using hyperspectral visible/near-infrared cameras (Hyperspec-VNIR-C) in Chikusei, Ibaraki, Japan, on July 29, 2014 [55]. This dataset comprises 19 classes and a size of 2517 × 2335 pixels. It comprises 128 spectral bands ranging from 363 to 1018 nm.

2) TD Dataset: Reflective Optics Spectrographic Image System (ROSIS) gathered the Pavia Centre (PC) dataset.[1] This dataset includes 1095 × 715 pixels. The collection includes 102 spectral bands from 430 to 860 nm and has nine classes.

[1]The PC dataset is publicly available and can be accessed in here.

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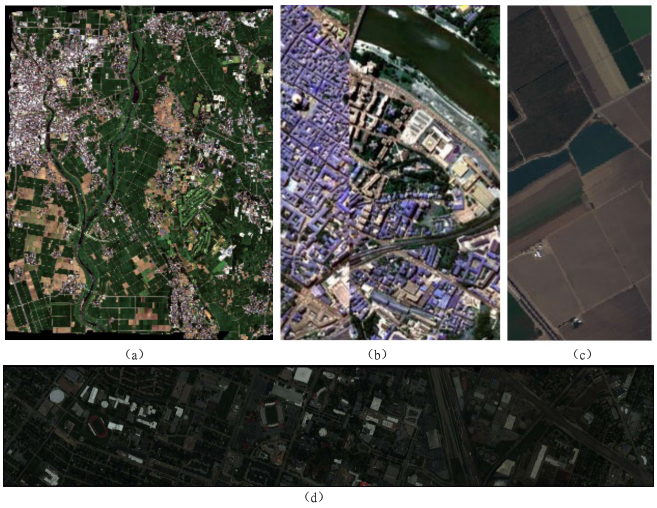


Fig. 4. False-color image. (a) Chikusei. (b) PC. (c) SA. (d) Houston.

TABLE I
LAND-COVER CLASSES OF THE CHIKUSEI DATASET WITH THE NUMBER OF SAMPLES FOR EACH CLASS

Class NO.	Class Name	Piexls
1	Water	2845
2	Bare soil(school)	2859
3	Bare soil(park)	286
4	Bare soil(farmland)	48525
5	Natural plants	730
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice filed(grown)	13369
10	Rice field(first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade(non-dark)	1220
14	Manmade(dark)	7664
15	Manmade(blue)	431
16	Manmade(red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	total	77592

California's Salinas Valley was captured by an AVIRIS sensor in the Salinas (SA) dataset.[2] This dataset includes 512 × 127 pixels with 224 bands. To account for water vapor absorption, 20 bands were removed, leaving 204 spectral bands from 400 to 2500 nm with 16 classes.

The Compact Airborne Spectrographic Imager (CASI) acquired the Houston dataset, covering the University of Houston campus and the surrounding urban region. This dataset boasts a high spatial resolution of 349 × 1905 pixels. It encompasses 15 classifications and comprises 144 spectral bands spanning from 380 to 1050 nm.[3]

[2]The SA dataset is publicly available and can be accessed in here.

[3]The data were provided by Prof. N. Y Okoya from the University of Tokyo and RIKEN AIP.

TABLE II
LAND-COVER CLASSES OF THE PC DATASET WITH THE NUMBER OF SAMPLES FOR EACH CLASS

Class NO.	Class Name	Pixels
1	Water	65971
2	Trees	7598
3	Asphalt	3090
4	Bricks	2685
5	Bitumen	6584
6	Tiles	9248
7	Shadows	7287
8	Meadows	42826
9	Bare oil	2863
	total	148152

TABLE III
LAND-COVER CLASSES OF THE SA DATASET WITH THE NUMBER OF SAMPLES FOR EACH CLASS

Class NO.	Class Name	Pixels
1	Brocoli_green_weeds 1	2009
2	Brocoli_green_weeds 2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
	total	54129

B. Experimental Setup

We selected different classification methods separately for adequate comparison in the method comparison experiments. These methods include a deep learning method called spectral–spatial 3-D CNN (SS CNN) [56], seven cross-domain FSL method DFSL [20], DCFSL [31] CMFSL [37], Gia-FSL [21], GCC-FSL [57], DMCM [58], and SSTFSL [46] in our conference version. In the experiments, for the supervised method SS CNN, we use only the labeled data in the TD as the only data for training. For the DCFSL, CMFSL, Gia-FSL, GCC-FSL, DMCM, and our SSTFSL method, we select five labeled TD samples per class for training. Chikusei for SD samples can learn more transferable knowledge. For the DFSL, we used Chikusei, Botswana, and Kennedy Space Center (KSC) to form the SD classes. In total, there are 46 classes, and they were randomly selected; 200 were labeled from SD.

We uniformly use the 9 × 9 neighborhood as the spatial size of the input cube. The training was performed via the Adam optimizer. The number of training iterations was set to 2000, which was sufficient to train the network. The learning

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TABLE IV
LAND-COVER CLASSES OF THE HOUSTON DATASET WITH THE NUMBER OF SAMPLES FOR EACH CLASS

Class NO.	Class Name	Pixels
1	Healthy Grass	1251
2	Stressed Grass	1254
3	Synthetic Grass	697
4	Tree	1244
5	Soil	1242
6	Water	325
7	Residential	1268
8	Commercial	1244
9	Road	1252
10	Highway	1227
11	Railway	1235
12	Parking Lot1	1234
13	Parking Lot2	469
14	Tennis Court	428
15	Running Track	660
	total	15011

rate was 0.001. We denote the number of classes in each set by C and the number of samples per class in the support set by N. Each set is a N-shot, C-way classification task. C is set to the class number of the target dataset (e.g., nine classes for the PC, 15 for the Houston, and 16 for the SA). For both source and target FSLs, N is set to 1, i.e., only one randomly selected labeled sample in each set is involved in method training.

We use the overall accuracy (OA), average accuracy (AA), and Kappa coefficients as the precision index in the experimental results. By comparing the classification result images with the ground-truth (GT) data, the confusion matrix $\mathbf{U}$ (a C $\times$ C matrix) can be calculated. Where C is the number of classes, $\mathbf{U}_{ii}$ denotes the number of samples where category J is recognized as i. OA denotes the probability that the classification result is consistent with the reference classification result, and its calculation formula is as follows:

$$\mathrm{OA} = \frac{\sum_{i=1}^{C} \mathbf{U}_{ii}}{N} \tag{14}$$

where N represents the total number of test samples. Unlike the overall classification accuracy, the AA indicates the mean value of the percentage of each class that is correctly classified. Its calculation formula is as follows:

$$\mathrm{AA} = \frac{\sum_{i=1}^{C} \left(\mathbf{U}_{ii} / \sum_{i=1}^{C} \mathbf{U}_{ij} \right)}{C}. \tag{15}$$

In some cases, there is a certain degree of uncertainty in the classification results, and the Kappa coefficient can be used to estimate the recognition accuracy taking into account the effect of uncertainty on the classification results. Its calculation formula is as follows:

$$\mathrm{Kappa} = \frac{\left(N \left(\sum_{i=1}^{C} \mathbf{U}_{\mathrm{ii}} \right) - \sum_{i=1}^{C} \left(\sum_{j=1}^{C} \mathbf{U}_{ij} \sum_{j=1}^{C} \mathbf{U}_{ji} \right) \right)}{\left(N^2 - \sum_{i=1}^{C} \left(\sum_{j=1}^{C} \mathbf{U}_{ij} \sum_{j=1}^{C} \mathbf{U}_{ji} \right) \right)}. \tag{16}$$

C. Comparison With Representative Baseline Methods

Tables V–VII report the detailed classification accuracies of different methods on the PC, SA, and Houston TD datasets. Intuitively, FSL methods (DFSL, DCFSL, CMFSL, Gia-FSL, GCC-FSL, DMCM, and SSTFSL) have a greater advantage over deep learning methods (SS CNN) when the labeled samples are small. This is due to the ability to learn from fewer samples that can be generalized to new categories based on meta-learning ideas. Deep learning methods, on the other hand, can only be limited by modeling the labeled samples. Thus, it is limited by the number of labeled samples available for training.

Among the FSL methods, DFSL is less effective as it fails to handle source and TD datasets within a unified framework. Conversely, DCFSL integrates the training processes across different domains into a united model, enhancing the model's generalization capabilities and improving classification performance. Both CMFSL and DMCM optimize decision boundaries and mitigate misclassification by focusing on metric types. However, they overlook the importance of sample quality and the intrinsic capacity of samples to convey transferable knowledge.

Gia-CFSL and GCC-FSL leverage graph structures to enhance generalization and feature extraction abilities, yet they do not utilize semantic information across different domains as a semantic signal for domain alignment. While SSTFSL incorporates semantic information, it only employs shallow semantic information, yielding limited results. The S^{3}CFSL method mitigates the influence of non-essential information on knowledge transfer through designed cooperation between feature extraction and feature denoising operations. The enhanced SEDA method effectively employs the semantic-aware strategy to facilitate domain alignment, addressing the challenges of domain shift and ultimately improving classification results.

Figs. 5–7 show the classification map obtained by each method on PC, SA, and Houston TD datasets. Predictably, SS CNNs struggle to handle the case of sample scarcity, resulting in the noisiest classification maps. Among the FSL methods, while the DFSL method performs poorly due to its difficulty in handling cross-domain datasets, the others demonstrate their respective effectiveness to varying degrees, yielding more desirable classification maps. However, the classification results of S^{3}CFSL are closer to the actual situation. This is because the other methods often overlook the high-quality processing of the samples themselves. The S^{3}CFSL method effectively captures both spatial and spectral features by integrating spectral and spatial channels during feature extraction. Additionally, it leverages semantic information to model these features, resulting in a classification map that is significantly improved in terms of accuracy and detail.

D. Ablation Experiments

We investigated the performance of S^{3}CFSL in classification by adding different modules to the method. Given that the fundamental architecture of our feature extractor is derived from the Transformer model, the unaltered ViT [5] was selected as the foundational feature extractor to validate the

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TABLE V
DIFFERENT CLASSIFICATION METHODS ON THE PC DATASET

CLASS NO.	SS CNN	DFSL	DCFSL	CMFSL	Gia-FSL	GCC-FSL	DMCM	SSTFSL	Ours
1	75.99	94.72	**99.75**	99.51	98.25	95.48	99.23	98.18	99.53
2	84.75	88.32	89.42	76.82	93.63	93.18	94.22	**96.66**	93.55
3	77.74	94.08	87.01	96.39	88.39	95.86	93.02	**98.91**	95.13
4	96.84	95.94	94.26	96.39	94.64	97.35	97.04	84.99	**98.23**
5	91.91	89.37	94.82	89.19	92.08	90.82	**96.11**	93.82	95.90
6	69.55	**96.25**	93.12	94.18	94.42	95.97	95.28	93.81	92.34
7	80.83	85.32	92.96	93.86	**94.03**	93.28	91.04	89.03	90.34
8	87.03	94.23	98.42	99.03	98.00	**99.57**	95.34	99.04	98.63
9	95.22	98.67	98.27	96.48	98.60	97.35	94.65	97.84	**99.33**
OA	83.31	96.76	97.12	96.96	97.44	98.45	98.33	97.85	**98.52**
AA	77.25	93.59	94.12	93.84	95.26	95.53	95.14	94.83	**95.77**
Kappa	0.8531	0.9376	0.9569	0.9548	0.9581	0.9716	0.9715	0.9603	**0.9728**

TABLE VI
DIFFERENT CLASSIFICATION METHODS ON THE SA DATASET

CLASS NO.	SS CNN	DFSL	DCFSL	CMFSL	Gia-FSL	GCC-FSL	DMCM	SSTFSL	Ours
1	82.62	82.71	**98.39**	82.80	94.41	87.08	85.36	84.42	83.69
2	96.80	**98.40**	92.11	96.62	98.31	95.30	95.43	97.27	96.25
3	98.21	93.66	99.60	**99.80**	72.81	98.41	95.80	95.84	94.33
4	97.82	96.49	96.78	**99.24**	93.02	92.32	93.66	97.53	98.67
5	97.35	**100.00**	97.73	97.73	**100.00**	98.44	**100.00**	**100.00**	**100.00**
6	85.31	92.30	95.10	95.10	**99.52**	95.10	96.23	95.10	98.03
7	88.33	86.47	57.28	76.99	88.29	80.87	86.84	**89.73**	87.56
8	60.58	**77.87**	68.09	55.55	68.26	77.68	68.33	77.01	77.52
9	54.95	72.52	53.92	67.42	77.08	75.88	**88.23**	78.78	78.30
10	54.34	50.96	77.47	68.05	**79.38**	57.33	75.32	64.28	69.48
11	76.18	72.86	84.91	82.35	98.02	85.76	95.62	82.92	**98.88**
12	72.04	66.18	77.23	58.50	78.07	**89.91**	83.12	83.38	79.44
13	85.96	75.78	50.88	60.00	90.86	88.42	**92.75**	91.92	88.76
14	99.19	92.71	98.38	98.79	98.40	**100.00**	93.22	97.97	98.73
15	**99.57**	96.61	98.52	98.73	85.01	97.04	95.31	94.08	98.55
16	81.45	81.41	**98.91**	85.05	86.51	90.87	96.32	82.93	93.11
OA	80.77	81.99	87.71	85.89	87.62	88.33	87.23	86.86	**88.79**
AA	83.46	83.71	83.09	88.75	90.11	90.04	89.35	88.68	**90.41**
Kappa	0.7914	0.8050	0.8018	0.8533	0.8625	0.8711	0.8674	0.8573	**0.8725**

ablation studies. Table VIII reports the effect of each module in classification.

1) Feature Extraction: We verified the necessity by experimenting with the feature extraction process. The comparison results are shown in Table VIII. We can find that using only the unaltered ViT [5] as the extraction network through removing the feature extraction process from our design will get an undesirable classification result. The rationale is that it is possible to efficiently extract comprehensive spatial and spectral features through SSDC. These features are further analyzed through the different channels of the co-embedded feature extraction. The spatial and spectral features in CSST interacts with the fused convolutional blocks through the cross-attention mechanism to obtain interactive spatial–spectral features and enhance fusion features for improving the final classification result.

2) Feature Denoising: From the experimental results in Table VIII, it can be seen that the classification results are further improved with the feature denoising operations and feature extraction. Feature denoising operations further obtain the essential features, reducing the effect of internal and external noise, which can promote the transferability of the information, which has an impact on the final result.

We also conducted ablation experiments for different filters in the feature denoising operations, as shown in Table IX, where it can be seen that the mean filter performs in general, and on the Houston dataset, even reduces the final method. This is because it does not keep the image details well and destroys the detailed parts of the features while denoising, thus making the features indistinguishable. The median filter performs better as a nonlinear filter, which keeps the sharp edges of the feature map and adds appropriately to the unconventional parts. Gaussian filtering is the best among them. Gaussian filtering gives a big weight to the near points and a small weight to the farther points, which can obtain more inherent feature. It is worth noting that denoised features are not enough by itself. Since

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TABLE VII
DIFFERENT CLASSIFICATION METHODS ON THE HOUSTON DATASET

CLASS NO.	SS CNN	DFSL	DCFSL	CMFSL	Gia-FSL	GCC-FSL	DMCM	SSTFSL	Ours
1	38.65	47.70	**66.13**	58.65	43.01	56.38	63.21F	63.26	57.14
2	42.79	19.77	40.61	42.79	61.58	38.34	**66.28**	43.26	50.22
3	52.51	32.51	58.62	68.10	81.90	77.16	**89.35**	80.60	79.54
4	47.10	81.82	**94.30**	80.93	74.56	77.10	78.21	69.57	77.43
5	86.25	73.65	69.45	76.18	84.28	86.25	88.02	87.96	**89.60**
6	64.96	**88.45**	64.96	71.20	72.80	73.92	83.54	72.91	75.73
7	86.02	53.35	77.70	74.84	69.98	81.28	83.21	**86.26**	80.69
8	70.34	56.42	70.34	62.29	52.49	**74.10**	70.32	66.69	63.47
9	55.36	43.58	70.55	68.44	62.06	58.69	**73.21**	49.33	70.23
10	47.98	41.35	25.49	47.98	55.64	52.22	63.24	62.44	**65.39**
11	59.96	66.71	65.51	57.59	60.53	**69.96**	67.23	62.75	68.68
12	36.56	37.40	27.13	38.21	33.84	36.56	40.25	**48.72**	44.36
13	95.53	85.71	**99.75**	97.50	97.00	98.00	93.31	99.35	98.36
14	98.00	62.57	76.35	83.44	88.49	84.63	95.36	85.40	**98.36**
15	96.34	95.12	96.34	**96.75**	80.49	**96.75**	94.32	95.37	80.67
OA	71.67	73.85	75.51	75.18	75.87	77.44	77.13	76.82	**77.63**
AA	73.28	75.24	76.82	77.89	76.13	79.76	78.44	78.43	**79.93**
Kappa	0.6857	0.7075	0.7358	0.7246	0.7390	0.7536	0.7454	0.7408	**0.7583**

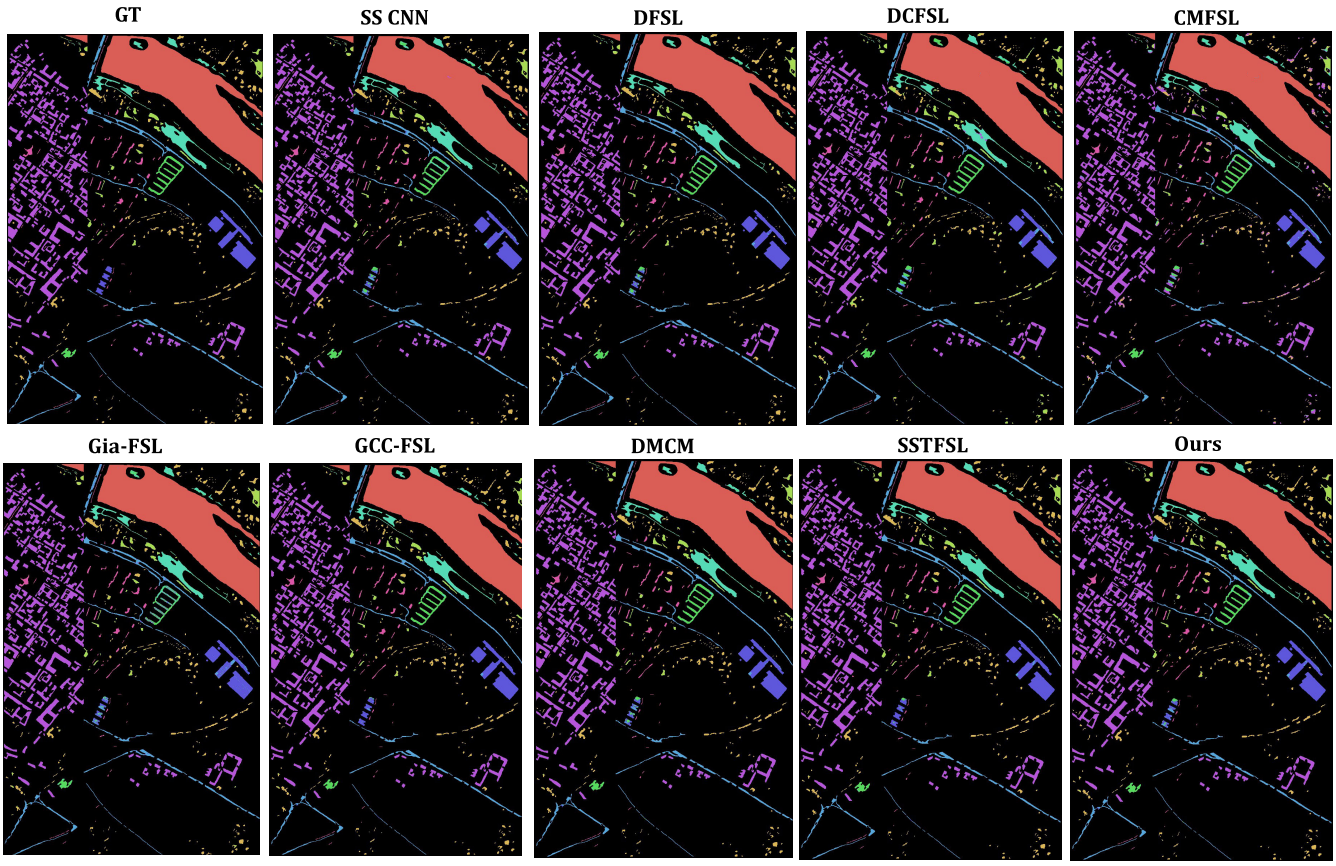


Fig. 5. Classification maps obtained by different methods on the PC dataset.

over-denoising may also remove useful information, we combine the denoised features with the original features in the denoising block rather than using the denoised features alone.

For a more intuitive presentation, we visualized the clustering of the features into classes, as shown in Fig. 8. It can be seen that the image without feature denoising and SEDM (first column) is the worst performer in all three datasets.

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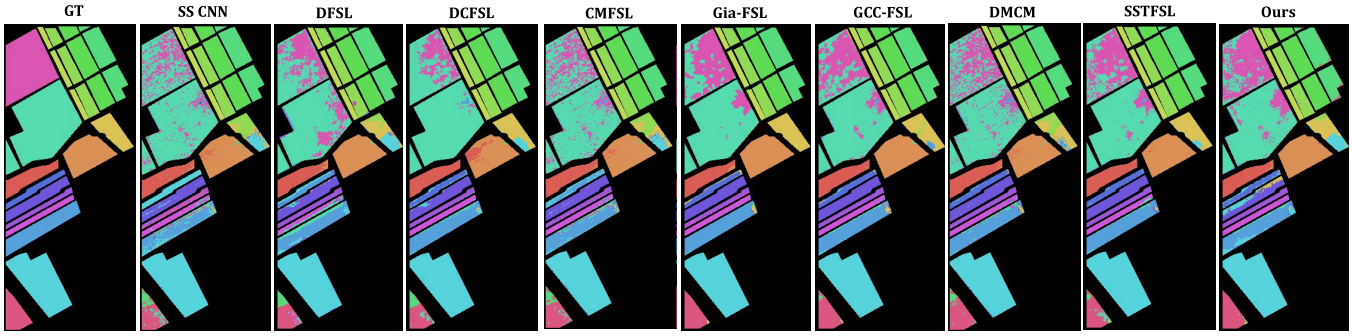


Fig. 6. Classification maps obtained by different methods on the SA dataset.

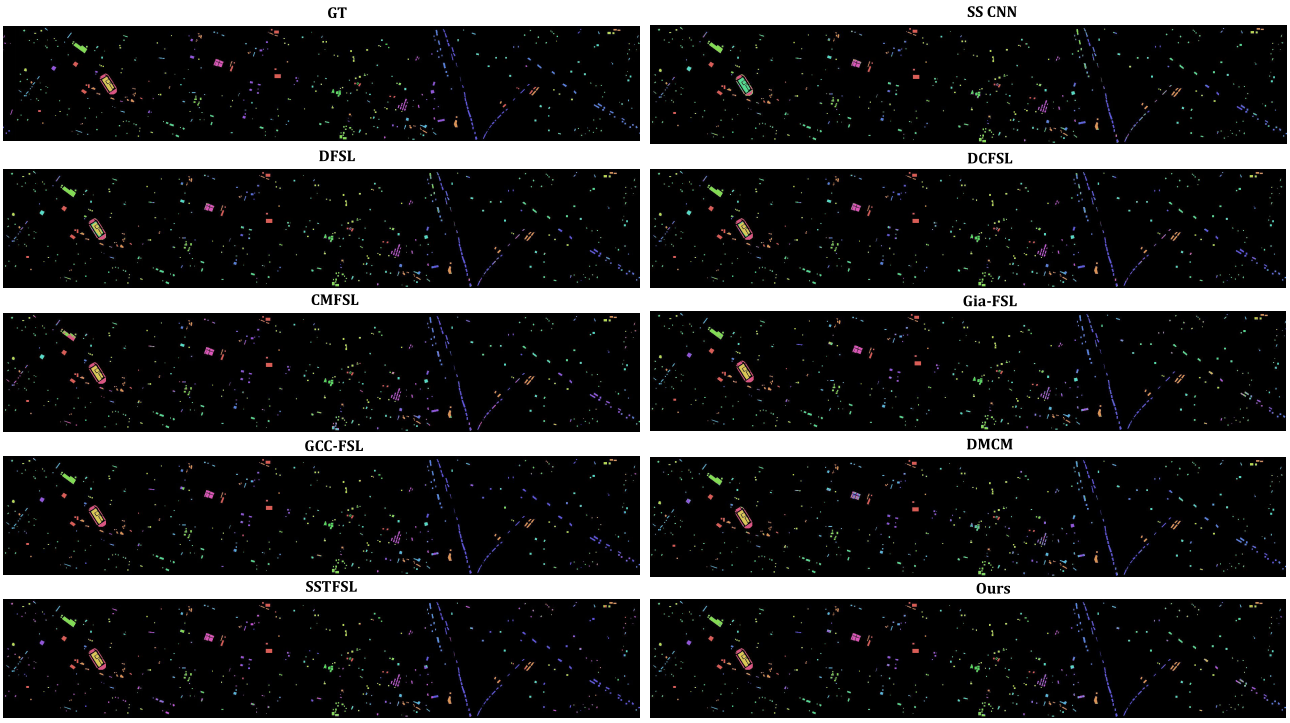


Fig. 7. Classification maps obtained by different methods on the Houston dataset.

TABLE VIII

ABLATION ANALYSIS OF THE S^{3}CFSL WITH A COMBINATION OF DIFFERENT MODULES FOR THE THREE DATASETS ON OA

No.	Feature Extraction	Feature Denoising	SEDA	PC	SA	Houston
1	✘	✘	✘	94.33	81.46	69.77
2	✘	✘	✓	96.24	87.57	74.83
3	✓	✘	✘	96.47	86.39	75.16
4	✓	✓	✘	97.25	87.44	75.69
5	✓	✓	✓	**98.52**	**88.79**	**77.63**

TABLE IX

ABLATION EXPERIMENTS ON FILTER SELECTION IN FEATURE DENOISING ON THREE DATASETS ON OA

Method	PC	SA	Houston
null	98.36	88.06	77.14
Mean filter	98.48	88.55	77.08
Median filter	98.43	88.74	77.51
Gaussian filter	**98.52**	**88.79**	**77.63**

Adding feature denoising (second column) gives better performance in all three datasets, especially in the dataset with similar classes (SA). This is because the feature denoising operations produce high-quality learnable features, enabling the network to better differentiate features from similar classes, thereby making the clustering graph clearer and more intuitive.

3) Semantic-Enhanced Domain Alignment: Table VIII shows that our SEDA usually gets an improvement in classification results due to its better facilitation of domain alignment. The method performance decreases when SEDA is missing. This is because SEDA utilizes a series of domain discriminators to identify public and unique domain

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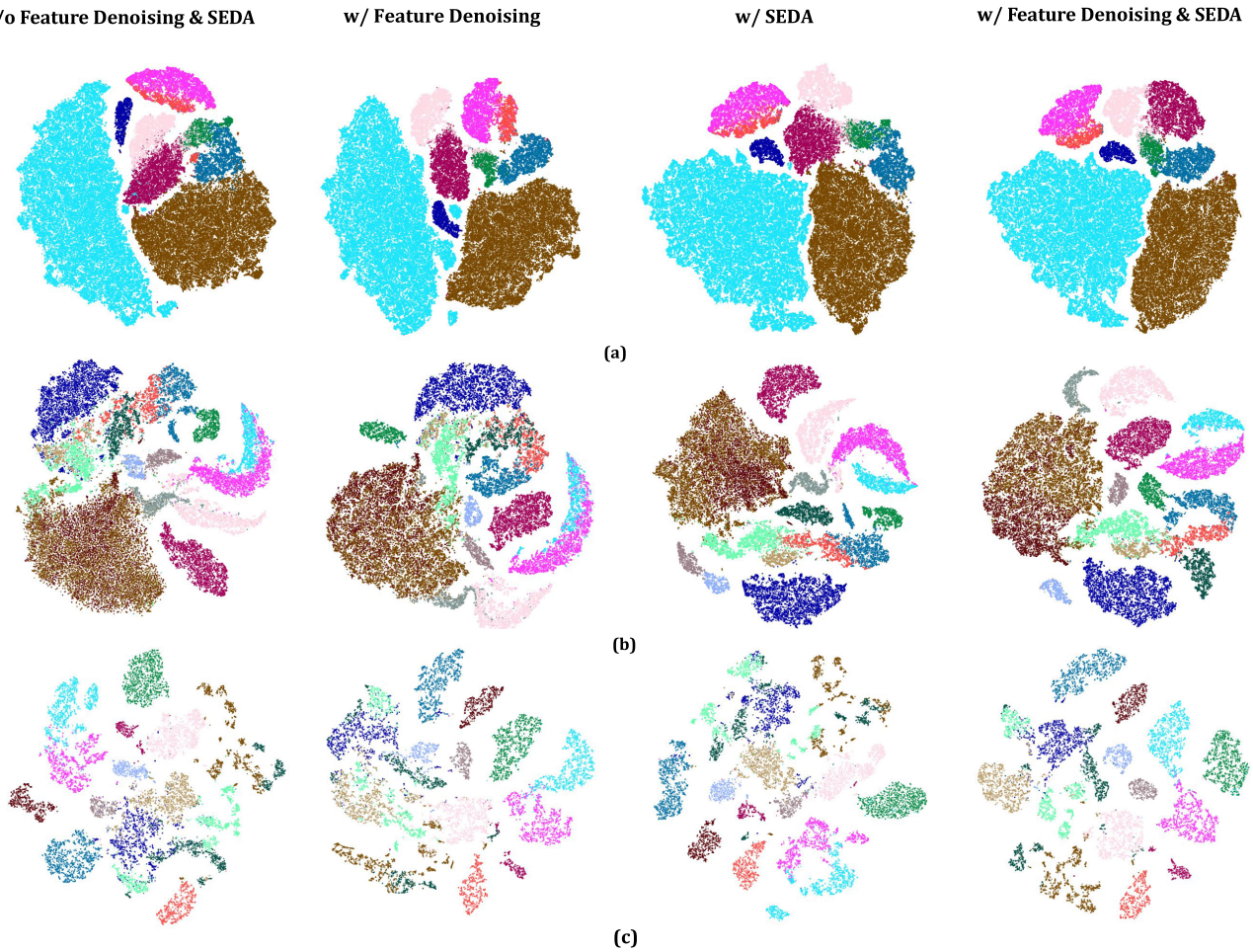


Fig. 8. Class separability of the proposed S^{3}CFSL using three datasets. (a) PC. (b) SA. (c) Houston. The first column denotes features without feature denoising and SEDA, the second column denotes features without SEDA, the third column denotes features without feature denoising, and the last column represents the features of our final model.

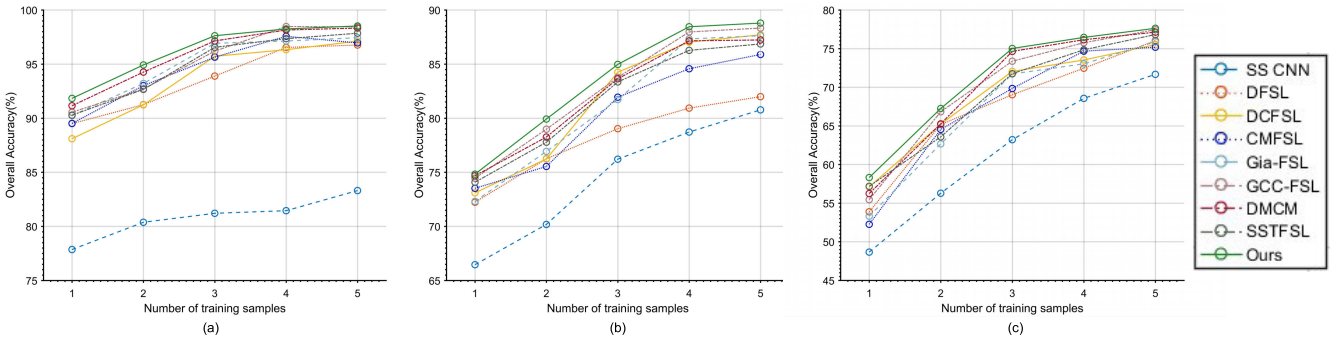


Fig. 9. Classification results with different numbers of support samples per class. (a) PC. (b) SA. (c) Houston.

information through a semantic-aware strategy and achieves domain alignment.

In Fig. 8, the feature clustering visualization of SEDM is also shown. It can be seen that the features are more clustered after the addition of SEDM (third column) compared to the features that were not added for feature denoising and SEDM (first column). This is due to the fact that our SEDM can be useful for facilitating domain alignment through the semantic-aware strategy, which results in a clearer and more efficient classification of samples in the TD. In the complete S^{3}CFSL (the last column), the cooperation of feature denoising and SEDM is even more powerful. The cooperation of the two can effectively improve the classification performance of the network under the HSI cross-domain few-shot task.

E. Adaptability Experiments on Different Numbers of Labeled Data

To ascertain the method's validity across varying sample sizes, we conducted experiments with the TD dataset by

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varying the number of samples from 1 to 5. Afterward, we compared the OAs of S^{3}CFSL on three datasets to DFSL, DCFSL, CMFSL, Gia-FSL, GCC-FSL, and SSTFSL under defined conditions. The comparison results are shown in Fig. 9.

S^{3}CFSL provides the highest classification accuracy in all datasets, showing strong adaptability to varying few labeled samples. In cross-domain few-shot tasks, noise features and sample selection bias invariably influence domain alignment, diminishing the accuracy of the proposed method. However, S^{3}CFSL effectively mitigates this problem using feature extraction, feature denoising operations, and SEDA. This further demonstrates that SSDC and CSST with feature denoising operations have significant advantages in a small number of FSL tasks. In addition, SEDA can effectively use a semantic-aware strategy to alleviate domain shift.

V. Conclusion

Currently, cross-domain FSL has created a boom in HSI field research. However, these research efforts have encountered shortcomings in preserving essential features in HSI, which in turn impedes the further transferability of meta-learning strategies. In addition, the lack of relevance of semantic information makes it difficult to cope with the problem of sample selection bias, which can easily lead to spurious modeling of different domains. In this article, we propose the S^{3}CFSL method to improve the performance of cross-domain few-shot HSI classification. It utilizes efficient feature extraction and feature denoising operations to obtain adequate essential information and aligns different domains through a semantic-aware strategy. In terms of feature extraction, we design SSDC and CSST to fully extract the interactive spatial–spectral features. This efficient feature extraction process is combined with feature denoising operations to facilitate transferable learning. In addition, under the effect of SEDA, the SD and TD are made closer in the feature space, which ultimately promotes distribution consistency. Experimental results on three different datasets show that our method performs better in HSI cross-domain few-shot classification.

However, practical applications demand a more efficient lightweight model capable of reflecting feature classification within a specific time frame. In the future, to better adapt to the challenges posed by evolving environments, we aim to develop a lightweight cross-domain hyperspectral few-shot classification network that can effectively handle noise interference.

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Mengxin Cao (Member, IEEE) received the BBM degree from the School of Information Management and Information System, Shandong First Medical University, Jinan, China, in 2021. She is currently pursuing the master's degree with the Faculty of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan.

She has authored or co-authored two papers in IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) and IEEE International Conference in Image Processing (ICIP). Her research interests include computer vision and image processing.



Xu Zhang is currently pursuing the M.E. degree with the School of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China.

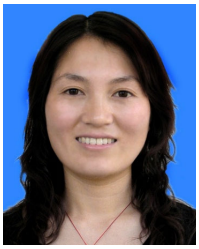
He has authored or co-authored two papers in *Knowledge-Based Systems* and the Conference on Information and Knowledge Management (CIKM). His research interests include artificial intelligence, spatio-temporal data mining, and computer vision.



Jinyong Cheng (Member, IEEE) received the M.S. degree in computer software and theory from Qingdao University, Qingdao, China, in 2005.

He is currently an Associate Professor with the School of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China. Until now, he has authored or co-authored more than 30 papers in conferences and journals. His research was sponsored by the Natural Science Foundation of Shandong province. His research interests include artificial intelligence, image processing, and biomedical information processing.

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Guixin Zhao received the Ph.D. degree from the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, China, in 2021.

From 2016 to 2017, she was a Visiting Scholar with Brunel University London, London, U.K. She is currently a Lecturer with the Faculty of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China. Her research interests include machine learning and hyperspectral image classification.



Wei Li (Senior Member, IEEE) received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2007, the M.S. degree in information science and technology from Sun Yat-sen University, Guangzhou, China, in 2009, and the Ph.D. degree in electrical and computer engineering from Mississippi State University, Starkville, MS, USA, in 2012.

Subsequently, he spent one year as a Post-Doctoral Researcher with the University of California at Davis, Davis, CA, USA. He is currently a Professor with the School of Information and Electronics, Beijing Institute of Technology, Beijing, China. He has authored or co-authored more than 150 peer-reviewed articles and 100 conference papers totally cited by 7500 times (Google Scholar). His research interests include hyperspectral image analysis, pattern recognition, and data compression.

Dr. Li received the JSTARS Best Reviewer Award in 2016 and TGRS Best Reviewer Award in 2020 from the IEEE Geoscience and Remote Sensing Society (GRSS), and the Outstanding Paper Award at the IEEE International Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (Whispers), in 2019. He is currently serving as an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), and IEEE SIGNAL PROCESSING LETTERS (SPL).



Xiangjun Dong received the Ph.D. degree in computer applications from Beijing Institute of Technology, Beijing, China, in 2005.

From 2007 to 2009, he worked as a Post-Doctoral Fellow with the School of Management and Economics, Beijing Institute of Technology. From 2009 to 2010, he was a Visiting Scholar at the University of Technology Sydney, Sydney, NSW, Australia. He is currently a Professor with the School of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China. He has authored or co-authored more than 100 journal/conference publications including *Artificial Intelligence*, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CYBERNETICS, *Pattern Recognition*, the Conference on Information and Knowledge Management (CIKM), and so on. His research interests include data mining, artificial intelligence, and big data.

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